An unconstrained Euclidean parametrisation of phylogenetic tree space

Xavier Didelot[1,2,*]

[1] School of Life Sciences, University of Warwick, Coventry, United Kingdom

[2] Department of Statistics, University of Warwick, Coventry, United Kingdom

[*] Corresponding author. Tel: 0044 (0)2476 572827. Email: xavier.didelot@warwick.ac.uk

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Abstract

Phylogenetic trees combine a discrete topology with continuous branch lengths, which complicates their representation in a fixed Euclidean space. We introduce a recursive contour-grafting construction that encodes both components using unconstrained real-valued coordinates. After fixing an order of the leaves, each new leaf is attached at a point selected by a contour traversal of the tree constructed so far. Moving a graft coordinate continuously along the contour permits changes of topology when internal branches contract to zero. For trees with n labelled leaves, we apply the construction to rooted ultrametric trees, rooted trees with unrestricted branch lengths, and unrooted trees, obtaining parametrisations of dimensions $n-1$, $2n-2$, and $2n-3$, respectively. We show that the resulting maps are surjective onto the corresponding spaces of binary trees with positive branch lengths and extend continuously to their completions. The maps are smooth within open contour cells but generally non-differentiable at boundaries where topology changes occur. We also determine the number of representations of a generic tree and calculate the Jacobian of each parametrisation. These results provide explicit change-of-variables formulae between Euclidean coordinates and the principal phylogenetic tree spaces, allowing probability distributions on tree space to be represented in unconstrained Euclidean coordinates.

1 Introduction

Phylogenetic trees combine a discrete topology with continuous branch lengths (Paradis 2006). This representation is natural, but it makes tree space difficult to explore using methods designed for vectors in a fixed Euclidean space. In particular, changing topology ordinarily requires explicit operations such as nearest-neighbour interchange or subtree pruning and regrafting, whose performance can depend strongly on the design and tuning of the proposals (Höhna and Drummond 2012; Zhang et al. 2020).

A geometric description of tree space is provided by Billera–Holmes–Vogtmann (BHV) space (Billera et al. 2001). Each tree topology is represented by an orthant of branch-length coordinates, and different orthants meet where one or more internal branches have length zero. This construction makes clear that topology changes can be viewed continuously: an edge belonging to one topology contracts before an edge belonging to another topology grows. The resulting space is nevertheless stratified, and computational methods that move between its orthants generally require specialised treatment (Dinh et al. 2017).

Here we develop a different representation in which both topology and branch lengths are encoded by a single vector of unconstrained real parameters. The construction is recursive. After fixing an order of the leaves, each new leaf is attached to the tree formed by the preceding leaves. Its graft point is selected by following a contour traversal of the current tree, so that continuous movement of a real-valued coordinate moves the graft point along branches and through vertices. In this way, changes of topology arise without introducing a separate discrete topology variable.

We apply this contour-grafting construction to rooted ultrametric trees, rooted trees with unrestricted branch lengths, and unrooted trees. Our aim is to describe the resulting parametrisations and their geometric and measure-theoretic properties, including their behaviour at topology boundaries and the relationship between Euclidean coordinates and standard local coordinates on tree space. The emphasis is on the parametrisations themselves rather than on any particular inference algorithm.

The remainder of the article is organised as follows. Section 2 introduces the recursive contour-grafting construction. Sections 3–5 apply it to rooted ultrametric, rooted unrestricted, and unrooted trees, respectively. Section 6 summarises and compares the three parametrisations. Finally, Section 7 provides numerical validation.

2 Recursive contour-grafting construction

All our parametrisations are constructed recursively using the same basic principle. We consider an arbitrary but fixed order for the leaves. The tree is constructed by adding each leaf one by one to the tree built so far. Let T_i denote the rooted tree formed by the first i leaves and let $L(T_i)$ be the sum of the branch lengths in the tree T_i. The tree T_1 consists only of the first leaf. When adding each leaf i beyond the first one, a graft point x_i is selected on or above T_{i-1} corresponding to the point where the new leaf is attached. This process is repeated for all leaves from the second one ($i = 2$) to the last one ($i = n$), each time adding to the tree T_{i-1} built previously.

The location of a graft point x_i is parametrised using a variable $a_i \in \mathbb{R}$. If $0 \leq a_i \leq 2L(T_{i-1})$ then a_i corresponds to the distance travelled along the contour traversal of T_{i-1} before reaching the graft point. Starting at the root, the traversal follows each branch once away from the root and once back towards it, visiting the first child of each internal node and then the second. Let $|T|$ denote the set of points on the tree T. The process of finding this graft point on a tree T using the contour traversal is denoted $\gamma_T : [0, 2L(T)] \to |T|$.

We further define the extended contour traversal as illustrated in Figure 1 and described below. If $a_i > 2L(T_{i-1})$, the grafting point is above the root of T_{i-1} by length $a_i - 2L(T_{i-1})$. Likewise, if $a_i < 0$ then the grafting point is above the root of T_{i-1} by length $-a_i$. This extended contour traversal is therefore a function $\Gamma_T : \mathbb{R} \to |T|^+$. The co-domain $|T|^+$ is obtained by adding to $|T|$ the points on a branch above the root at a distance $d \in [0, \infty)$, with zero identified with the root. The extended contour traversal is defined as:

$$\Gamma_T(a) = \begin{cases} \text{point above root by } -a & \text{if } a < 0 \\ \gamma_T(a) & \text{if } 0 \leq a \leq 2L(T) \\ \text{point above root by } a - 2L(T) & \text{if } a > 2L(T) \end{cases} \tag{1}$$

At the graft point $x_i = \Gamma_{T_{i-1}}(a_i)$, a new node is created with first child equal to what was previously below that point, and second child equal to the new branch of length ℓ_i leading to the leaf i. This convention also defines the ordering of the children used in subsequent contour traversals. If x_i lies above the current root, the segment joining the old root to x_i is added to the tree and x_i becomes the new root. If x_i coincides with an existing vertex, the operation is interpreted as the corresponding limiting construction with a zero-length branch. Thus the construction takes values in the completion

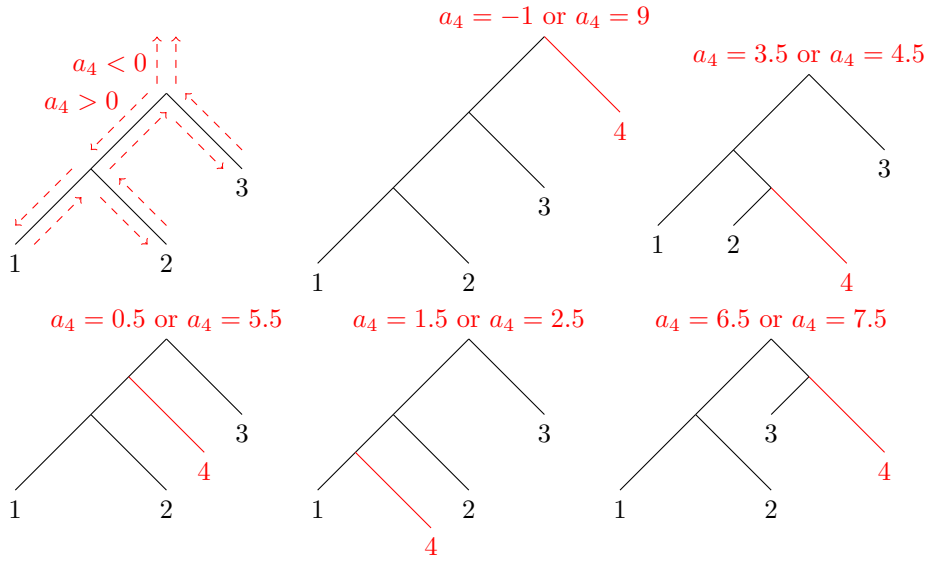


Figure 1: Illustration of the extended contour traversal. The initial tree is shown in black, with three leaves labeled 1, 2 and 3 and all four branches of length 1. Leaf 4 is added with the value of a_4 determining the graft point as shown and the new terminal branch shown in red.

of binary tree space, while almost every parameter vector produces a binary tree with strictly positive branch lengths.

The extended contour traversal is continuous because $\gamma_T(0) = \gamma_T(2(L(T))) = \text{root}(T)$. Every point in the interior of an edge of T is visited exactly twice, once in each traversal direction. A point at distance $d > 0$ above the root also has exactly two preimages, namely $-d$ and $2L(T) + d$. Vertices are exceptional because they may be visited more than twice; these exceptional points form the boundaries between contour regions. We can deduce the following recursive formula for the sum of the branch lengths $L(T_i)$:

$$L(T_i) = \begin{cases} 0 & \text{if } i = 1 \\ L(T_{i-1}) + \ell_i + \max(0, a_i - 2L(T_{i-1})) + \max(0, -a_i) & \text{otherwise} \end{cases} \tag{2}$$

3 Rooted ultrametric trees

We consider phylogenetic trees that are binary, ultrametric and rooted, with n labeled leaves. There are $n-1$ internal nodes including the root. The topology of the tree is discrete and there are $(2n-3)!! = (2n-2)!/(2^{n-1}(n-1)!)$ unique possibilities (Felsenstein 1978). The branch lengths are continuous but not independent from each other. Since the tree is ultrametric, the branch lengths are constrained so that the number of free parameters corresponds to the $n-1$ values for the heights of the internal nodes relative to the contemporaneous leaves.

The previous section introduced the recursive principle for constructing trees using the extended contour traversal for grafting, with parameters $(a_i)_{i=2,\dots,n} \in \mathbb{R}^{n-1}$ used in Equation 1. This construction produces an ultrametric rooted tree because, once a graft point has been selected, the length of the newly created terminal branch is uniquely determined by the requirement that the new leaf is contemporaneous with the existing leaves. In particular, the dimensionality of the parameters is $n-1$ which is the dimensionality needed to specify the $n-1$ heights of the internal nodes. For a point x on an ultrametric tree, let $h_T(x)$ denote its distance to any descendant leaf. For a point at distance d above the current root r, its distance is $h_T(x) = h_T(r) + d$.

Let $\mathcal{C}_n^1 = \mathbb{R}^{n-1}$ denote the parameter space. Let $\mathcal{T}_n^1$ denote the space of rooted binary ultrametric trees with n labelled leaves and strictly positive branch lengths. Let $\overline{\mathcal{T}}_n^1$ denote its completion obtained by allowing zero-length branches and identifying trees that define the same labelled metric tree after contraction of all zero-length edges. An element of $\mathcal{C}_n^1$ is the vector $C = (a_i)_{i=2,\dots,n}$, and the construction defines a unique tree in $\overline{\mathcal{T}}_n^1$. It therefore defines a function $\Phi_n^1 : \mathcal{C}_n^1 \longrightarrow \overline{\mathcal{T}}_n^1$ which is detailed in Algorithm 1. Almost every value of C produces a tree in $\mathcal{T}_n^1$; the exceptional values correspond to grafting at vertices, including the contour endpoints. A small example of applying Algorithm 1 is shown in Figure 2.

The function Φ_n^1 is surjective onto $\mathcal{T}_n^1$, meaning that for any phylogenetic tree $T \in \mathcal{T}_n^1$ there is at least one parametrisation C such that $\Phi_n^1(C) = T$. To see this, consider any $T \in \mathcal{T}_n^1$, remove leaf n and its pendant branch, and suppress its resulting degree-two parent. Repeating this operation produces trees with successively fewer leaves. Reversing these steps reconstructs T using the proposed grafting procedure, because every required grafting location is reached by the extended contour traversal and ultrametricity uniquely determines the length of each newly added terminal branch. However, the function Φ_n^1 is not injective as there are more than one C such that $\Phi_n^1(C) = T$. This is a consequence of each grafting point being visited twice by the contour traversal: once as the algorithm goes down

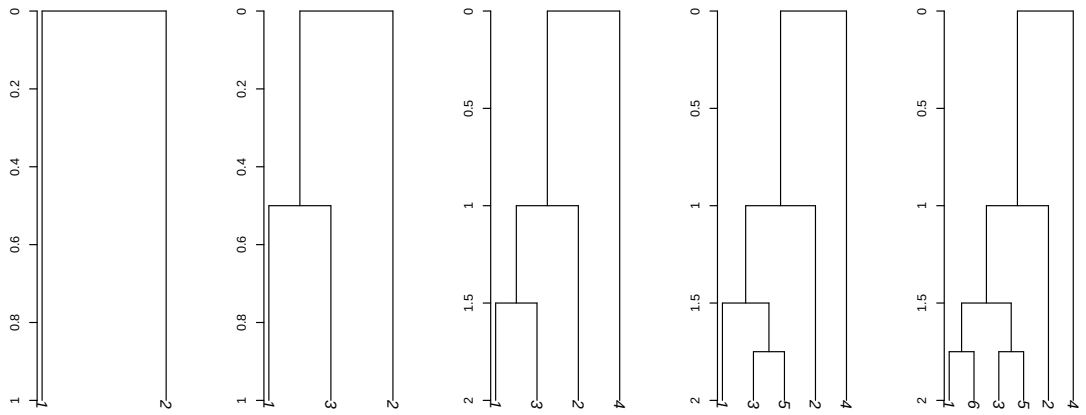


Figure 2: An example of construction of an ultrametric tree using as input $n = 6$ and $(a_i)_{i=2,\dots,n} = (-1, 1.5, -1, 2.75, 2.25)$.

the branch and then again as it goes back up the branch. Likewise, a grafting point above the root by length x can be obtained by both $a_i = -x$ and $a_i = 2L(T_{i-1}) + x$. Each grafting point is therefore represented exactly twice if we exclude the possibility of branches of length zero. For an ultrametric rooted binary tree T with strictly positive branch lengths, there are therefore 2^{n-1} distinct values of C such that $\Phi_n^1(C) = T$. The duplication of grafting coordinates is needed to achieve continuity: it allows the grafting location to move continuously through the endpoints of the contour and across changes in tree topology, without an explicit jump between discrete topological states.

Algorithm 1 Construction of ultrametric rooted trees

Require: Ordered leaves $1, \dots, n$ and parameters $C = (a_i)_{i=2,\dots,n} \in \mathbb{R}^{n-1}$
Ensure: An ultrametric rooted phylogenetic tree T_n

Let T_1 consist of the single leaf 1, which is also its root
$L(T_1) \leftarrow 0$
for $i = 2, \dots, n$ **do**
    Find the grafting point $x_i = \Gamma_{T_{i-1}}(a_i)$ using Equation 1
    If x_i lies above the current root, let x_i be the new root
    Insert a new internal node at x_i, splitting its branch if necessary
    Make the subtree previously below x_i the first child of this node
    Attach leaf i as its second child by a branch of length $\ell_i = h_{T_{i-1}}(x_i)$
    Denote the resulting tree by T_i
    $L(T_i) \leftarrow L(T_{i-1}) + \ell_i + \max(0, -a_i) + \max(0, a_i - 2L(T_{i-1}))$
end for
return T_n

Equip $\overline{\mathcal{T}}_n^1$ with the restriction of the completed rooted Billera–Holmes–Vogtmann metric (Billera et al.

2001), including pendant-edge lengths. We now show that $\Phi_n^1 : \mathcal{C}_n^1 \to \overline{\mathcal{T}}_n^1$ is continuous. Within each region of parameter space in which the selected contour segments and traversal directions are fixed, all node heights and branch lengths vary continuously with the parameters. When a grafting location passes through an internal node, the separating internal branch contracts continuously to zero before the adjacent topology is entered. The two topologies therefore meet at the same boundary tree in $\overline{\mathcal{T}}_n^1$. When the grafting location passes through a leaf, the relevant pendant segment tends continuously to zero and the two contour directions likewise meet at the same boundary tree. Similarly, the segment added above the root tends continuously to zero as a_i approaches either endpoint of the contour, namely $a_i = 0$ or $a_i = 2L(T_{i-1})$. Applying this argument inductively at each grafting step establishes the continuity of Φ_n^1.

Within each binary topology stratum, the creation-ordered internal-node heights provide local coordinates. Densities on tree space are understood with respect to counting measure on topologies and Lebesgue measure in these local height coordinates.

Finally we calculate the Jacobian of $\Phi_n^1 : \mathcal{C}_n^1 \to \overline{\mathcal{T}}_n^1$ within its open binary topology strata. Let $\boldsymbol{h} = (h_2, ..., h_n)$ where h_i is the height of the internal node created when leaf i is grafted. Within a fixed contour region, $h_i = h_{T_{i-1}}(\Gamma_{T_{i-1}}(a_i))$. The new height depends only on $a_2, ..., a_i$, not on later parameters. Moreover,

$$\frac{\partial h_i}{\partial a_i} \in \{-1, +1\}$$

because height changes at unit rate along every fixed contour segment and along the branch above the root. Therefore the Jacobian matrix is triangular, with diagonal entries $+1$ or -1, and

$$\left| \det \frac{\partial(h_2, ..., h_n)}{\partial(a_2, ..., a_n)} \right| = 1 \tag{3}$$

This identity holds away from contour-region boundaries, which form a set of Lebesgue measure zero in $\mathcal{C}_n^1$.

4 Rooted trees with unrestricted branch lengths

We consider phylogenetic trees that are binary and rooted, with n labeled leaves. There are $n - 1$ internal nodes including the root. The topology of the tree is discrete and there are $(2n - 3)!! = (2n - 2)!/(2^{n-1}(n - 1)!)$ unique possibilities (Felsenstein 1978). The branch lengths are continuous,

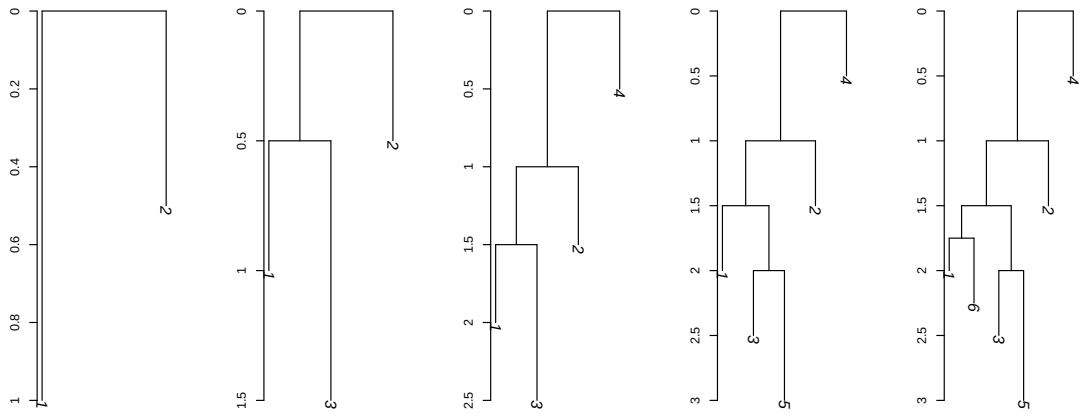


Figure 3: An example of construction of a rooted using as input $n = 6$, $(a_i)_{i=2,...,n} = (-1, 1.5, -1, 3, 2.25)$ and $(b_i) = i = 2, ..., n = \log(0.5, 1, 0.5, 1, 0.5)$.

and there are $2n - 2$ branch lengths taking values in $\mathbb{R}_{>0}$ (one for each leaf and non-root internal node). Our construction is similar to the one above for ultrametric trees. The main difference is that there is now another parameter $b_i \in \mathbb{R}$ for each leaf beyond the first one, that represents the length $\exp(b_i)$ of the branch separating the graft point (determined by a_i) and the newly added leaf i.

Let $\mathcal{C}_n^2 = \mathbb{R}^{2n-2}$ denote the parameter space. Let $\mathcal{T}_n^2$ denote the space of rooted binary phylogenetic trees with n labelled leaves and strictly positive branch lengths. Let $\overline{\mathcal{T}}_n^2$ denote its completion obtained by allowing zero-length branches and identifying trees that define the same labelled metric tree after contraction of all zero-length edges. An element of $\mathcal{C}_n^2$ is the vector $C = ((a_i, b_i))_{i=2,...,n}$, and the construction defines a function $\Phi_n^2 : \mathcal{C}_n^2 \longrightarrow \overline{\mathcal{T}}_n^2$ which is detailed in Algorithm 2. By the same arguments as in the ultrametric case, this map is continuous and is surjective onto $\mathcal{T}_n^2$. Almost every parameter vector produces a tree in $\mathcal{T}_n^2$, and every tree in $\mathcal{T}_n^2$ has exactly 2^{n-1} preimages. A small example of applying Algorithm 2 is shown in Figure 3.

As previously, we calculate the Jacobian of $\Phi_n^2 : \mathcal{C}_n^2 \to \overline{\mathcal{T}}_n^2$ within its open binary topology strata. We consider this Jacobian locally, within a region of $\mathcal{C}_n^2$ where the resulting topology, the selected contour segment for each grafting point, and its traversal direction are fixed. Let $\boldsymbol{\ell}^{(i-1)}$ denote the vector of branch lengths in T_{i-1}. If leaf i is grafted in the interior of a branch of length ℓ_j, that branch is

Algorithm 2 Construction of rooted phylogenetic trees

Require: Ordered leaves $1, \dots, n$ and parameters $C = ((a_i, b_i))_{i=2,\dots,n} \in \mathbb{R}^{2n-2}$
Ensure: A rooted phylogenetic tree T_n
  Let T_1 consist of the single leaf 1, which is also its root
  $L(T_1) \leftarrow 0$
  for $i = 2, \dots, n$ **do**
    Find the grafting point $x_i = \Gamma_{T_{i-1}}(a_i)$ using Equation 1
    If x_i lies above the current root, let x_i be the new root
    Insert a new internal node at x_i, splitting its branch if necessary
    Make the subtree previously below x_i the first child of this node
    Attach leaf i as its second child by a branch of length $\exp(b_i)$
    Denote the resulting tree by T_i
    $L(T_i) \leftarrow L(T_{i-1}) + \exp(b_i) + \max(0, -a_i) + \max(0, a_i - 2L(T_{i-1}))$
  end for
  return T_n

replaced by two branches of lengths

$$u_i(\boldsymbol{\ell}^{(i-1)}, a_i) \quad \text{and} \quad \ell_j - u_i(\boldsymbol{\ell}^{(i-1)}, a_i)$$

and a terminal branch of length $\exp(b_i)$ is added. Along any fixed segment and direction of the contour traversal, u_i is affine in a_i, with

$$\frac{\partial u_i}{\partial a_i} \in \{-1, 1\}$$

After reordering the branch-length coordinates, the absolute Jacobian determinant of this grafting step is therefore

$$\left| \det \frac{\partial \boldsymbol{\ell}^{(i)}}{\partial(\boldsymbol{\ell}^{(i-1)}, a_i, b_i)} \right| = \left| \frac{\partial u_i}{\partial a_i} \right| \frac{\partial \exp(b_i)}{\partial b_i} = \exp(b_i)$$

With a suitable ordering of the branch-length coordinates, the derivative of the full recursive construction is block triangular. Consequently, the determinants contributed by successive grafting steps multiply. The same result holds when the grafting point lies above the root, since the new root branch has derivative -1 or $+1$ with respect to a_i. Applying recursively gives, for every C in which the grafting points do not coincide with a node or an endpoint of the contour:

$$\left| \det \frac{\partial \boldsymbol{\ell}(\Phi_n^2(C))}{\partial C} \right| = \prod_{i=2}^{n} \exp(b_i) = \exp\left(\sum_{i=2}^{n} b_i \right) \tag{4}$$

This identity holds away from contour-region boundaries, which form a set of Lebesgue measure zero in $\mathcal{C}_n^2$.

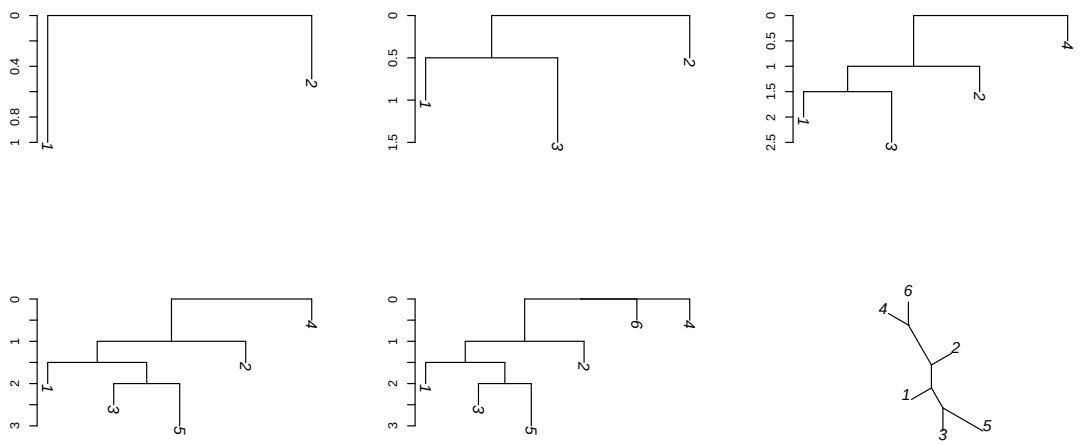


Figure 4: An example of construction of an unrooted tree using as input $n = 6$, $(a_i)_{i=2,\ldots,n-1} = (-1, 1.5, -1, 3)$, $(b_i)_{i=2,\ldots,n-1} = \log(0.5, 1, 0.5, 1)$ and $c = \log(0.5)$.

5 Unrooted trees

We now consider unrooted binary phylogenetic trees with $n \geq 3$ labelled leaves. Such a tree has $n-2$ internal nodes, $(2n-5)!!$ possible topologies, and $2n-3$ branch lengths. We obtain an unconstrained parametrisation of the same dimension by distinguishing leaf n and applying the rooted construction of Section 4 to the first $n-1$ leaves.

Let $C = ((a_i, b_i)_{i=2,\ldots,n-1}, c) \in \mathcal{C}_n^3 = \mathbb{R}^{2n-3}$. First construct a rooted tree R on leaves $1, \ldots, n-1$ using Algorithm 2, with leaf 1 as the initial leaf and parameters $(a_i, b_i)_{i=2,\ldots,n-1}$. Attach leaf n directly to the root of R by a branch of length $\exp(c)$, and then forget the root designation. The former root has degree three and therefore becomes an ordinary internal node of the resulting unrooted tree. This construction is detailed in Algorithm 3. A small example of applying Algorithm 3 is shown in Figure 4.

Algorithm 3 Parametrisation of unrooted phylogenetic trees

Require: Ordered leaves $1, \ldots, n$ and parameters $C = ((a_i, b_i)_{i=2,\ldots,n-1}, c) \in \mathbb{R}^{2n-3}$
Ensure: An unrooted phylogenetic tree T_n
Apply Algorithm 2 to leaves $1, \ldots, n-1$ and parameters $(a_i, b_i)_{i=2,\ldots,n-1}$ to obtain the rooted tree R_{n-1}
Attach leaf n to the root of R_{n-1} by a branch of length $\exp(c)$
Forget the root designation and denote the resulting unrooted tree by T_n
return T_n

Let $\mathcal{T}_n^3$ denote the space of unrooted binary phylogenetic trees with n labelled leaves and strictly positive branch lengths, and let $\overline{\mathcal{T}}_n^3$ denote its completion obtained by allowing zero-length branches and identifying trees that define the same labelled metric tree after contraction of all zero-length edges. The construction defines a function $\Phi_n^3 : \mathcal{C}_n^3 \longrightarrow \overline{\mathcal{T}}_n^3$. Almost every parameter vector produces a tree in $\mathcal{T}_n^3$; the exceptional values correspond to grafting at vertices, including contour endpoints.

The map is surjective onto $\mathcal{T}_n^3$. To see this, consider any $T \in \mathcal{T}_n^3$. Remove leaf n and its pendant branch, and regard the resulting degree-two vertex as the root of the remaining tree on leaves $1, \ldots, n-1$. This is a rooted binary phylogenetic tree and can therefore be represented by the construction of Section 4. Setting c to the logarithm of the removed pendant-branch length reconstructs T.

Every tree in $\mathcal{T}_n^3$ has exactly 2^{n-2} preimages. Indeed, the rooted tree on leaves $1, \ldots, n-1$ has 2^{n-2} representations by the result of Section 4, while the length of the branch leading to leaf n uniquely determines c. Thus the generic fibre cardinality of Φ_n^3 is 2^{n-2}.

Equip $\overline{\mathcal{T}}_n^3$ with the completed unrooted Billera–Holmes–Vogtmann metric, including pendant-edge lengths. The map Φ_n^3 is continuous. This follows from the continuity of the rooted map Φ_{n-1}^2, together with the continuity of attaching a pendant edge of length $\exp(c)$ at the root and then forgetting the root designation. At topology boundaries, zero-length edges are identified in $\overline{\mathcal{T}}_n^3$ in the same way as in the rooted construction.

Finally, we calculate the Jacobian within the open binary topology strata. Use the $2n-3$ unrooted branch lengths as local coordinates. The rooted construction on leaves $1, \ldots, n-1$ contributes $\exp\left(\sum_{i=2}^{n-1} b_i\right)$ to the absolute Jacobian, and the additional pendant branch has length $\exp(c)$ and derivative $\exp(c)$. With a suitable ordering of the coordinates, the full derivative is block triangular, and hence

$$\left|\det \frac{\partial \boldsymbol{\ell}(\Phi_n^3(C))}{\partial C}\right| = \exp\left(c + \sum_{i=2}^{n-1} b_i\right) \tag{5}$$

almost everywhere.

6 Summary and comparison

The three constructions above provide unconstrained Euclidean parametrisations of the principal phylogenetic tree spaces considered in this article. In every case, the dimension of the parameter space

Tree space	Parameters	Dimensions	Multiplicity	Absolute Jacobian
Rooted ultrametric	$(a_i)_{i=2,\dots,n}$	$n-1$	2^{n-1}	1
Rooted unrestricted	$(a_i, b_i)_{i=2,\dots,n}$	$2n-2$	2^{n-1}	$\exp\left(\sum_{i=2}^{n} b_i\right)$
Unrooted	$((a_i, b_i)_{i=2,\dots,n-1}, c)$	$2n-3$	2^{n-2}	$\exp\left(c + \sum_{i=2}^{n-1} b_i\right)$

Table 1: Summary of the three parametrisations.

equals the dimension of each open binary topology stratum. The main properties are summarised in Table 1.

All three maps are surjective onto the corresponding space of binary trees with strictly positive branch lengths and extend continuously to the completed tree space obtained by allowing zero-length edges. Their parameter spaces are divided into open contour cells on which the selected contour segments and traversal directions are fixed. Within each such cell, the resulting topology is fixed and the map is smooth. The boundaries between cells correspond to grafting at vertices or contour endpoints. At these boundaries an edge contracts to zero, so adjacent binary topologies meet at the same tree in the completed space. The maps are therefore continuous globally, although they need not be differentiable across contour-cell boundaries.

The rooted ultrametric construction is the simplest case. Each added leaf requires only its graft coordinate, because ultrametricity uniquely determines the new terminal-branch length. The creation-ordered internal-node heights depend triangularly on the graft coordinates, with diagonal derivatives equal to $+1$ or -1, giving unit absolute Jacobian. Allowing unrestricted branch lengths introduces one additional log-length parameter b_i at every grafting step. Exponentiation of these parameters contributes the factors $\exp(b_i)$ to the Jacobian. The unrooted construction applies the unrestricted rooted construction to leaves $1, \dots, n-1$, attaches the distinguished leaf n to the root with length $\exp(c)$, and then forgets the root designation. It therefore removes one graft coordinate relative to the rooted case and has the minimal dimension $2n-3$.

The multiple representations arise from the contour traversal. For a generic rooted tree, every graft point has two contour coordinates, so the $n-1$ successive grafts give 2^{n-1} representations. In the unrooted construction, only the rooted subtree on $n-1$ leaves is constructed by contour grafting, giving 2^{n-2} representations; the parameter c is uniquely determined by the length of the pendant edge leading to leaf n. These counts can change on the measure-zero set of boundary trees, where grafting points coincide with vertices.

The change-of-variables formula has the same form in all three cases. Let Φ denote one of the three maps, let m be its generic fibre cardinality, and let $J(C)$ be its absolute Jacobian from Table 1. If p_T is a density on the corresponding tree space, with respect to counting measure on topologies and Lebesgue measure in the stated local coordinates, then the natural symmetric pullback assigning equal mass to all generic representations has density

$$p_C(C) = \frac{1}{m}\, p_T(\Phi(C))\, J(C) \tag{6}$$

almost everywhere. Thus the three parametrisations differ in dimension, fibre multiplicity, and Jacobian, but share the same underlying contour-grafting mechanism and the same continuous, piecewise-smooth geometric structure.

7 Numerical validation

We consider rooted trees, such that all of the $(2n-3)!!$ topologies are equally likely and the branch lengths ℓ_i follow an Exponential distribution with mean θ, so that:

$$p_T(T) = \frac{1}{(2n-3)!!} \prod_{i=1}^{2n-2} \theta^{-1} \exp\left(-\frac{\ell_i}{\theta}\right) \tag{7}$$

To sample from this distribution, we use the adaptive Metropolis sampler (Vihola 2011). This algorithm uses a multivariate normal distribution as the proposal kernel $q(C'|C)$ which is symmetric ($q(C'|C) = q(C|C')$), so that the acceptance probability is simply $\min(1, p_C(C')/p_C(C))$ calculated using Equation **??**. The mean of the proposal kernel is the current value C and its covariance matrix is tuned to achieve an acceptance rate of 0.234 (Andrieu and Thoms 2008). We take $n = 5$ as a small example, with $\theta = 2$ as the mean of the distribution of branch lengths. The sampler was run for 2×10^5 iterations, with the first half discarded as burn-in and the remainder sampled every 100 iterations to produce a sample of size 1000. This run took less than a minute on a standard laptop. Figure 5 shows that the trees sampled by the MCMC are distributed as expected under Equation 7. All $(2n-3)!! = 105$ topologies have been sampled, the distribution of the sum of branch lengths and of a single branch length are as expected, and a majority-rule consensus tree looks completely star-like. The effective sample size was greater than 200 for individual branch lengths as computed using the R package coda (Plummer et al. 2006). The approximate effective sample size of the tree topology was also greater than 200 as computed by the R package rwty (Warren et al. 2017).

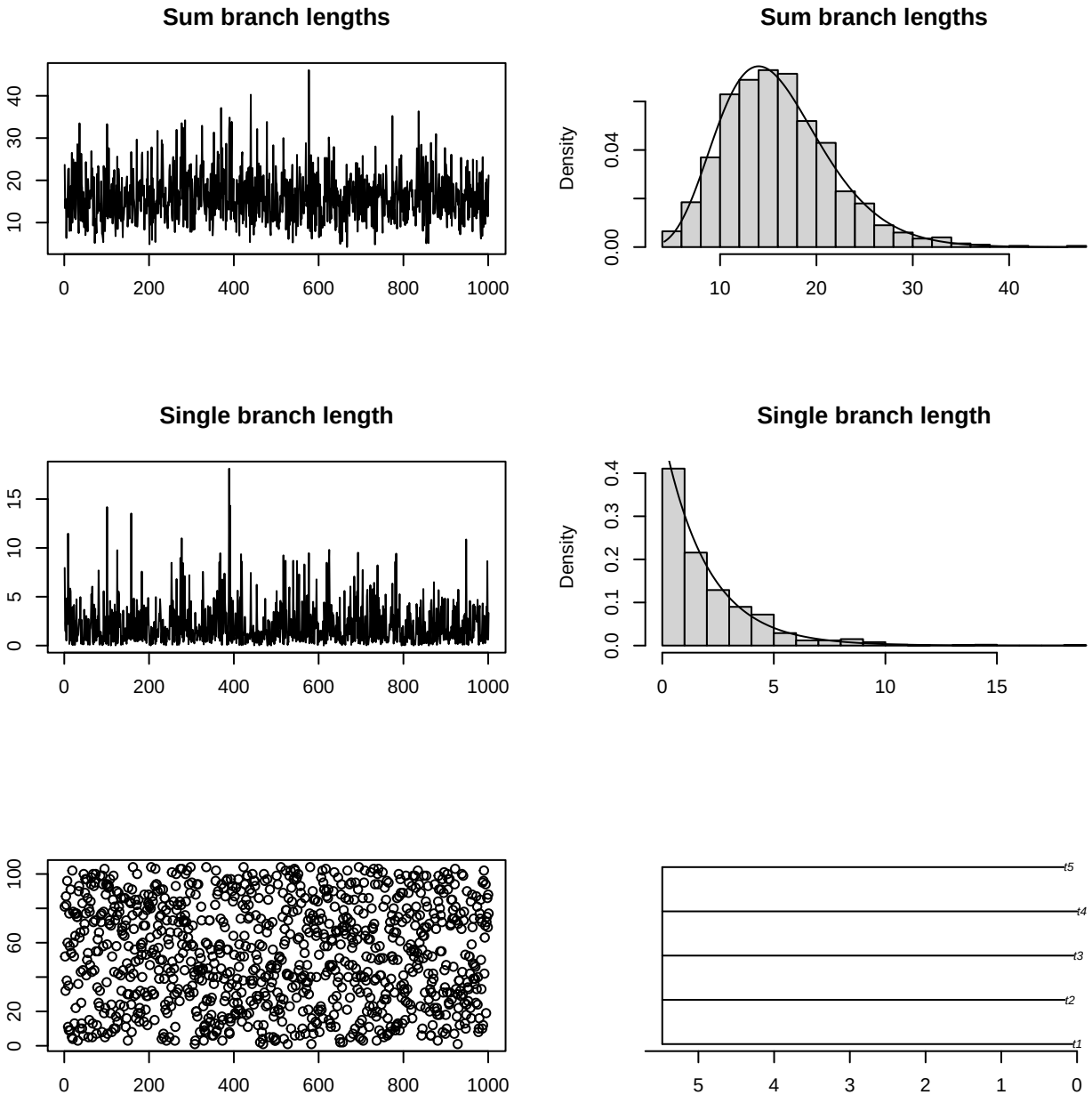


Figure 5: MCMC run. (A) Trace of sum of branch lengths. (B) Distribution of sum of branch lengths, sampled (histogram) and expected (line). (C) Trace of a single branch length. (D) Distribution of a single branch length, sampled (histogram) and expected (line). (E) Trace of topology. (F) Majority-rule consensus tree.

8 Discussion

The aim of Bayesian phylogenetic inference is to reconstruct a tree given some genetic data using Bayesian statistics to capture the posterior distribution of the tree (Nascimento et al. 2017). This is usually done using a Markov Chain Monte-Carlo (MCMC) to generate samples from this posterior distribution (Gilks et al. 1996). The MCMC needs to propose updates to both the branch lengths and the topology, with the latter using topology-changing moves such as nearest-neighbor interchange (NNI) or subtree pruning and regrafting (SPR). However, these moves are not easy to optimise (Höhna and Drummond 2012; Zhang et al. 2020). Conversely, many promising MCMC methods have not been applied in Bayesian phylogenetics because they can only be applied to continuous parameters. This includes in particular some adaptive methods where proposal parameters are learned during sampling (Andrieu and Thoms 2008), as well as gradient-based approaches such as the Langevin and Hamiltonian methods (Roberts and Rosenthal 1998; Neal 2011; Betancourt 2017). For example the dynamic Hamiltonian Monte Carlo (HMC) method implemented in Stan (Carpenter et al. 2017; Betancourt 2017). After observing data D made of n aligned genetic sequences of length l, the distribution becomes the posterior with density $p_C(C|D) \propto p_C(C)p(D|T = \Phi_n(C))$where the last term is the likelihood of the data given the phylogenetic tree $T = \Phi_n(C)$ obtained using Algorithm 2. This likelihood can be calculated efficiently using the Felsenstein pruning algorithm (Felsenstein 1981) which is used in most phylogenetic inference tools. Many substitution models can be used to calculate this likelihood (Nascimento et al. 2017) and here we use the simplest Jukes-Cantor model (Jukes and Cantor 1969) which avoids having to infer additional parameters.

References

Andrieu, C., Thoms, J., 2008. A tutorial on adaptive MCMC. Statistics and Computing 18, 343–373. doi:10.1007/s11222-008-9110-y.

Betancourt, M., 2017. A Conceptual Introduction to Hamiltonian Monte Carlo. arxiv arXiv:1701.02434.

Billera, L.J., Holmes, S.P., Vogtmann, K., 2001. Geometry of the Space of Phylogenetic Trees. Advances in Applied Mathematics 27, 733–767. doi:10.1006/aama.2001.0759.

Carpenter, B., Gelman, A., Hoffman, M.D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M.A., Guo, J., Li, P., Riddell, A., 2017. Stan: A probabilistic programming language. Journal of Statistical Software 76. doi:10.18637/jss.v076.i01.

Dinh, V., Bilge, A., Zhang, C., Matsen, F.A., 2017. Probabilistic Path Hamiltonian Monte Carlo. doi:10.48550/arXiv.1702.07814, arXiv:1702.07814.

Felsenstein, J., 1978. The Number of Evolutionary Trees. Systematic Zoology 27, 27. doi:10.2307/2412810.

Felsenstein, J., 1981. Evolutionary trees from DNA sequences: A maximum likelihood approach. Journal of Molecular Evolution 17, 368–376. doi:10.1007/BF01734359.

Gilks, W.R., Richardson, S., Spiegelhalter, D.J., 1996. Markov Chain Monte Carlo in Practice. volume 39. Chapman & Hall/CRC. doi:10.2307/1271145.

Höhna, S., Drummond, A.J., 2012. Guided Tree Topology Proposals for Bayesian Phylogenetic Inference. Systematic Biology 61, 1–11. doi:10.1093/sysbio/syr074.

Jukes, T., Cantor, C., 1969. Evolution of protein molecules, in: Mammalian Protein Metabolism. Academic Press, pp. 21–132.

Nascimento, F.F., Reis, M.D., Yang, Z., 2017. A biologist's guide to Bayesian phylogenetic analysis. Nature Ecology and Evolution 1, 1446–1454. doi:10.1038/s41559-017-0280-x.

Neal, R.M., 2011. MCMC Using Hamiltonian Dynamics. doi:10.1201/b10905, arXiv:1206.1901.

Paradis, E., 2006. Analysis of Phylogenetics and Evolution with R. Springer.

Plummer, M., Best, N., Cowles, K., Vines, K., 2006. CODA: Convergence diagnosis and output analysis for MCMC. R News 6, 7–11. doi:10.1159/000323281, arXiv:1011.1669v3.

Roberts, G.O., Rosenthal, J.S., 1998. Optimal Scaling of Discrete Approximations to Langevin Diffusions. Journal of the Royal Statistical Society Series B: Statistical Methodology 60, 255–268. doi:10.1111/1467-9868.00123.

Vihola, M., 2011. On the stability and ergodicity of adaptive scaling Metropolis algorithms. Stochastic Processes and their Applications 121, 2839–2860. doi:10.1016/j.spa.2011.08.006.

Warren, D.L., Geneva, A.J., Lanfear, R., 2017. RWTY (R We There Yet): An R package for examining convergence of Bayesian phylogenetic analyses. Molecular Biology and Evolution , msw279doi:10.1093/molbev/msw279.

Zhang, C., Huelsenbeck, J.P., Ronquist, F., 2020. Using Parsimony-Guided Tree Proposals to Accelerate Convergence in Bayesian Phylogenetic Inference. Systematic Biology 69, 1016–1032. doi:10.1093/sysbio/syaa002.